

3315. Construct the Minimum Bitwise Array II

Solved 

Medium

 Topics

 Companies

 Hint

You are given an array `nums` consisting of `n` **prime** integers.

You need to construct an array `ans` of length `n`, such that, for each index `i`, the bitwise **OR** of `ans[i]` and `ans[i] + 1` is equal to `nums[i]`, i.e. `ans[i] OR (ans[i] + 1) == nums[i]`.

Additionally, you must **minimize** each value of `ans[i]` in the resulting array.

If it is *not possible* to find such a value for `ans[i]` that satisfies the **condition**, then set `ans[i] = -1`.

Example 1:

Input: `nums = [2,3,5,7]`

Output: `[-1,1,4,3]`

Explanation:

- For `i = 0`, as there is no value for `ans[0]` that satisfies `ans[0] OR (ans[0] + 1) = 2`, so `ans[0] = -1`.
- For `i = 1`, the smallest `ans[1]` that satisfies `ans[1] OR (ans[1] + 1) = 3` is `1`, because `1 OR (1 + 1) = 3`.
- For `i = 2`, the smallest `ans[2]` that satisfies `ans[2] OR (ans[2] + 1) = 5` is `4`, because `4 OR (4 + 1) = 5`.
- For `i = 3`, the smallest `ans[3]` that satisfies `ans[3] OR (ans[3] + 1) = 7` is `3`, because `3 OR (3 + 1) = 7`.

Example 2:

Input: `nums = [11, 13, 31]`

Output: `[9, 12, 15]`

Explanation:

- For `i = 0`, the smallest `ans[0]` that satisfies `ans[0] OR (ans[0] + 1) = 11` is `9`, because `9 OR (9 + 1) = 11`.
- For `i = 1`, the smallest `ans[1]` that satisfies `ans[1] OR (ans[1] + 1) = 13` is `12`, because `12 OR (12 + 1) = 13`.
- For `i = 2`, the smallest `ans[2]` that satisfies `ans[2] OR (ans[2] + 1) = 31` is `15`, because `15 OR (15 + 1) = 31`.

Constraints:

- `1 <= nums.length <= 100`
- `2 <= nums[i] <= 109`
- `nums[i]` is a prime number.

Python:

class Solution:

```
def minBitwiseArray(self, nums: List[int]) -> List[int]:
    ans = []
    for n in nums:
        if n != 2:
            ans.append(n - ((n + 1) & (-n - 1)) // 2)
        else:
            ans.append(-1)
    return ans
```

JavaScript:

```
var minBitwiseArray = function(nums) {
    const ans = []
    for(let i = 0; i < nums.length; i++) {
        let n = nums[i]
        if(n != 2) ans[i] = n - ((n + 1) & (-n - 1)) / 2
        else ans[i] = -1
    }
}
```

```
    return ans  
};
```

Java:

```
class Solution {  
    public int[] minBitwiseArray(List<Integer> nums) {  
        int ans[] = new int[nums.size()];  
        for(int i = 0; i < nums.size(); i++) {  
            int n = nums.get(i);  
            if(n != 2) ans[i] = n - ((n + 1) & (-n - 1)) / 2;  
            else ans[i] = -1;  
        }  
        return ans;  
    }  
}
```